

Hoyt Field

Occupancy Counter

City of Cambridge, MA

Hoyt Field · MIT 6.900 · Team 5

Prepared for City of Cambridge



8 sensor nodes · 3 facilities · Battery-powered

Project Overview

A wireless, battery-powered system that automatically counts people entering and exiting park facilities: giving Cambridge a live view of how Hoyt Field is being used, without any cameras or personal data.



3 Facilities Monitored

Basketball, Baseball, Kiddie
Playground



LoRa + WiFi Communication

Child sensors transmit IR crossing
events to a parent ESP32 gateway,
which uploads data over park Wi-Fi



Battery-Powered Child Nodes

Child nodes are powered by 3 D-cell
batteries and designed for 6 months
of field operation

8

Sensor Nodes

3

Park Facilities

6

Months battery life

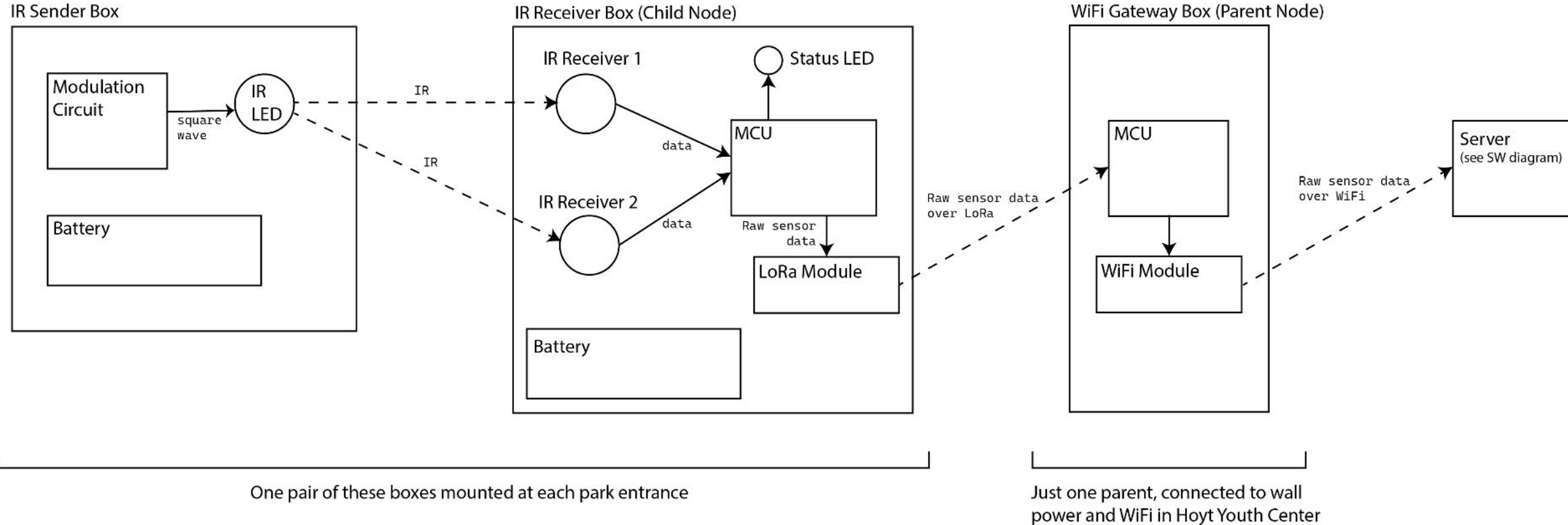
0

Line Power Required for
Child Nodes

0

Personal Data Collected

How the System Works



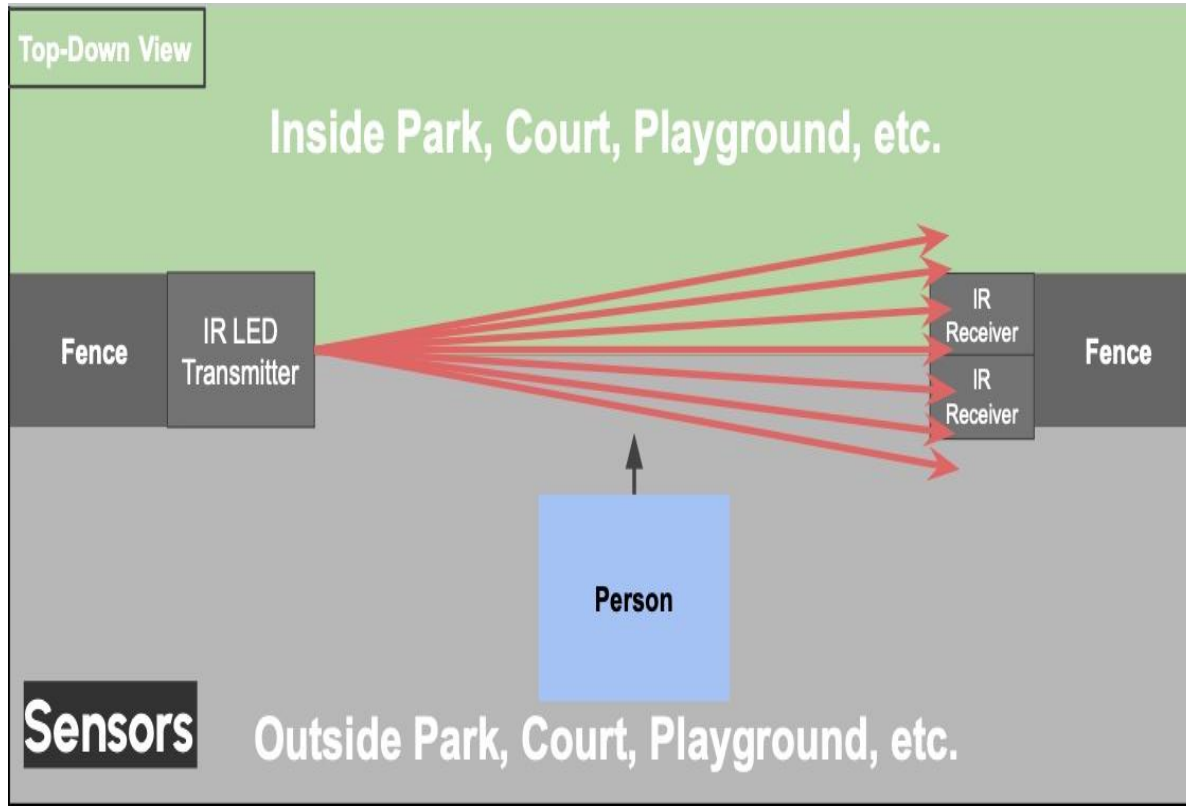
Node Locations

8 child nodes placed at every entrance to 3 facilities · 1 parent node at Youth Center (Wi-Fi)



Priority 1	Baseball Field	4
Nodes: F1, F2, F3, F4		
Priority 2	Basketball Courts	2
Nodes: B1, B2		
Priority 3	Kiddie Playground	2
Nodes: K1, K2		

How the Sensor Works



Key Design:

Two staggered IR receivers infer direction of travel.

1

IR Beam Emitter

An IR LED transmitter emits a wide-angle infrared beam across the gate opening, similar to older TV remote controls

2

Person Interrupts Beam

When someone walks through, they break the beam. The two-receiver sequence determines direction (IN vs OUT)

3

Delta Recorded

The MCU in the child node records ± 1 delta and transmits via LoRa to the parent node within seconds

4

Occupancy Count Updated on Website

The parent node then uploads the data to the website, where current occupancy and trends can be seen.

The Hardware

Weatherproof | Battery-powered | No permanent installation required



ESP32-C3 Microcontroller

MCU handles sensor ISRs, LoRa, and SD logging. It automatically light-sleeps when there are no crossing events.

LoRa Module

Child nodes communicate over LoRa to the parent gateway so no Wi-Fi needed at the sensor

D-cell Batteries + Buck Reg

3 D-Cell batteries provides 4.5V regulated to 3.3V with on-board reverse polarity protection

3D-Printed PETG Enclosure

Designed for Cambridge winters:
UV-resistant material, waterproof sealant, securely bolted enclosure

MicroSD Fault Logging

All events and faults are written to SD card for analysis even if network is unavailable

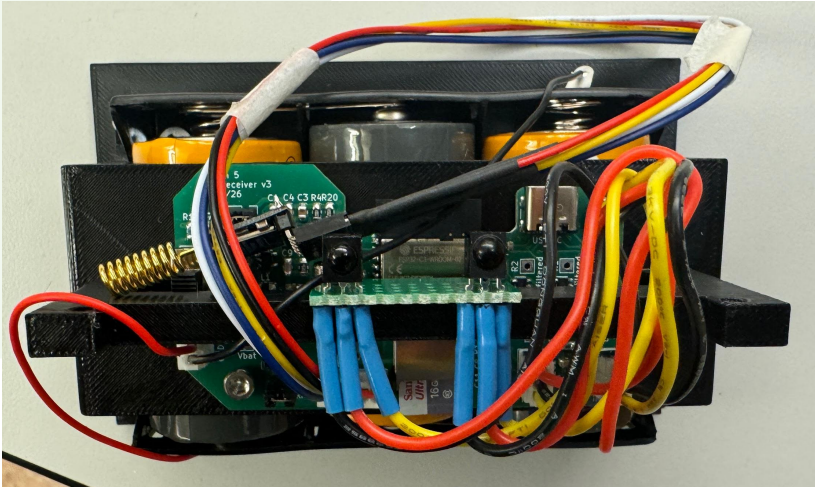
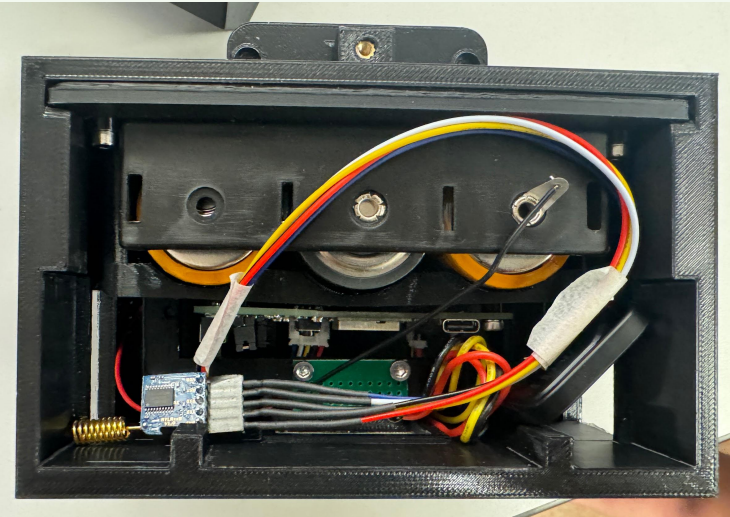
\$100 per Entry

Including material cost, device construction labor, and installation data

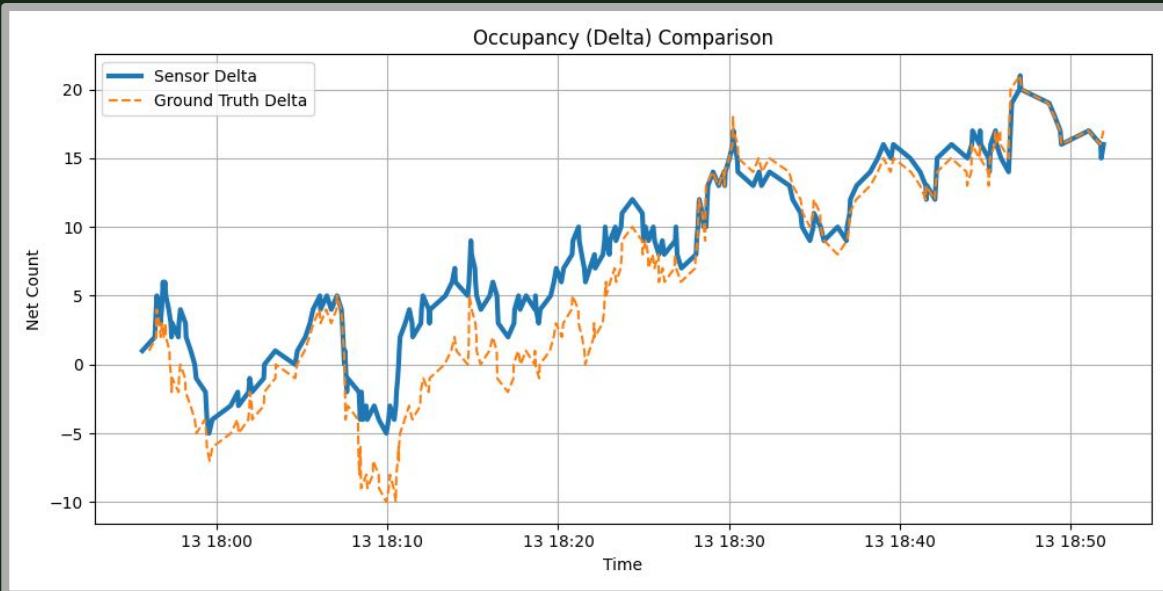
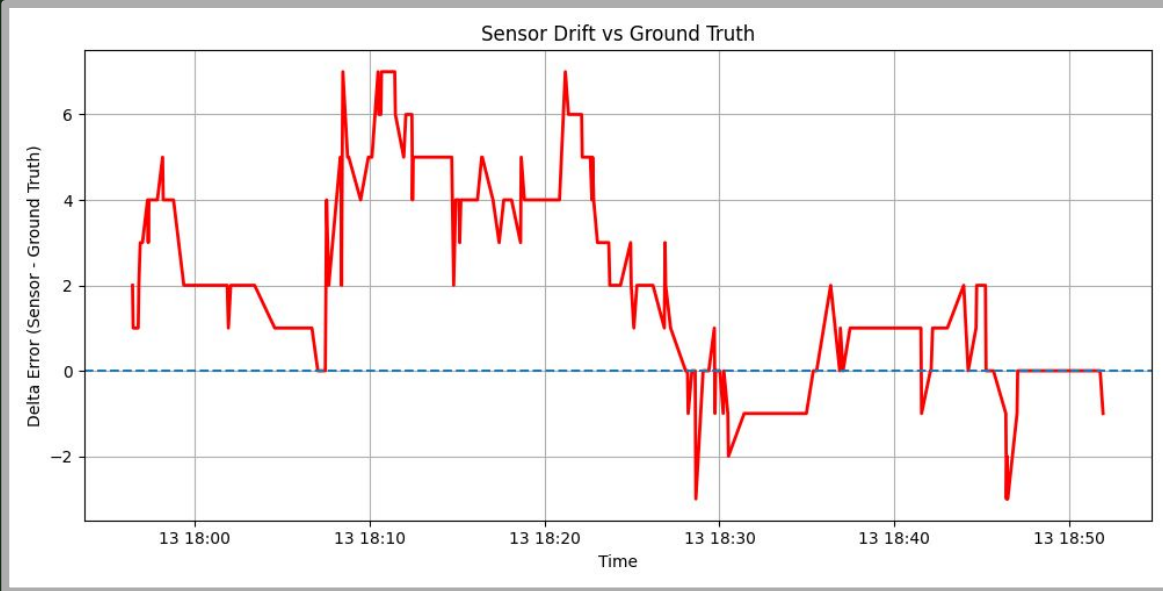
Bill of Materials & Cost of Goods

\$100

Item	Cost
Transmitter PCB	\$8.96
Transmitter Enclosure	\$10.39
Receiver PCB	\$22.53
Receiver Enclosure	\$14.00
Batteries	\$6.00
Assembly	\$25.00
Installation	\$15.00
Total	\$101.88



Sensor Testing + Deployed Hardware



Website

